SOFTWARE ENGINEERING – LEVEL 3 MINI-PROJECT

DELIVERABLE 2

Software Requirements Specification, Initial Validation Specification and Project Plan

Online Exam Portal – Web-Based Examination, Question Bank and Result Management System

Field	Details
Project ID	OEP-2026-13
Group / Team	Team 13 (G-13)
Course	UE24CS341A – Software Engineering, V Semester
Development Methodology	Agile (Scrum, 6 sprints)
Technology Stack	JavaScript – MERN Stack (MongoDB, Express.js, React.js, Node.js)
Customer / Stakeholder	Department Examination Cell / Course Faculty

Team Members

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Version 1.0 — Finalized for Deliverable 2

1. SRS – Required Structure and Coverage

This section completes the assignment-specific SRS structure table from the original Deliverable 2 and maps each required component to the corresponding section in this final document.

Section	Required content	Final D2 coverage
1. Introduction	Purpose, intended audience, product scope, references	Sections 2.2–2.4 and Appendix D
2. Overall Description	Product perspective, functions, user classes, environment, constraints, assumptions	Section 3
3. External Interface Requirements	User, software and communication interfaces	Section 4
4. Analysis Models	Use case and other relevant analysis models	Section 5
5. System Features	Feature description, priority, stimulus/response, detailed FRs	Section 6
6. Nonfunctional Requirements	Performance, safety, security, quality attributes, business rules	Section 7
7. Other Requirements	Database, data retention, deployment, localization and reuse considerations	Section 8
Appendix A	Glossary	Appendix A
Appendix B	Field layouts and report requirements	Appendix B
Appendix C	Requirement traceability matrix	Appendix C
Validation Specification	Planned validation methods and evidence	Section 9
Project Plan	Work packages, ownership, effort and outputs	Section 10
Tool Stack	Development, testing, CI/CD, containerization and design tools	Section 11

Alignment note: The final version preserves the project idea, target users, scope, objectives, technology stack and ten functional features established in Deliverable 1. The additions below complete the SRS structure without introducing AI proctoring, subjective-answer evaluation, ERP integration, native mobile applications, or live surveillance.

2. Introduction

2.1 Purpose

The purpose of the Online Exam Portal is to provide a digital platform for conducting examinations. Administrators/faculty can create categorized question banks, configure and schedule exams, and view performance reports. Students can authenticate, view available exams, attempt them within a fixed time limit, and view results after automated evaluation.

2.2 Intended Audience

- Project developers responsible for implementing the MERN-based system.
- Project testers responsible for functional, security, performance, usability and reliability validation.
- Project faculty/evaluators reviewing the requirements and planned evidence.
- Examination-cell/faculty stakeholders reviewing whether the proposed workflow meets the intended examination-management needs.

2.3 Product Scope

In Scope	Out of Scope
Student and administrator registration/login and role-based access.	AI/webcam-based online proctoring.
Question bank creation, editing, categorization and disabling.	Automated evaluation of descriptive/subjective answers.
Exam creation, configuration and scheduling.	Integration with external university examination/ERP systems.
Question randomization/shuffling for attempts.	Native Android/iOS mobile application.
Timed online examination with navigation, marked-for-review and auto-save.	Live audio/video invigilation and surveillance.
Automatic submission on time expiry.	—
Automatic evaluation and scoring of objective questions.	—
Result generation, student history/performance views and administrator reports.	—

2.4 References

- Deliverable 1 – Synopsis / Project Proposal, Online Exam Portal, OEP-2026-13.
- Deliverable 2 / SRS and project-flow requirements provided for the Software Engineering – Level 3 mini-project, 2026.
- This final Deliverable 2 is the controlled requirements baseline for subsequent architecture, implementation and testing work.

3. Overall Description

3.1 Product Perspective

The Online Exam Portal is a standalone web-based examination management system. It digitizes the examination workflow from question-bank preparation and exam scheduling to timed student attempts, automatic evaluation, result generation and performance reporting. The proposed implementation uses a React.js frontend, Node.js + Express.js backend, MongoDB database and JWT-based authentication.

3.2 Product Functions

- Register and authenticate students and administrators.
- Enforce role-based access between student and administrative functions.
- Create, edit, categorize and disable multiple-choice questions.
- Create and schedule examinations using duration, availability window and question-selection criteria.
- Randomly select and shuffle questions for individual student attempts.
- Present a timed examination interface with question navigation and marked-for-review state.
- Auto-save responses and automatically submit attempts when time expires.
- Evaluate objective questions and calculate/store results.
- Provide students with attempt history, scores and performance information.
- Provide administrators with exam-wise and student-wise reports.

3.3 User Classes and Characteristics

User class	Privileges	Characteristics
Student	Browse assigned exams, attempt exams, view own results/history/performance.	Non-technical user; requires simple, clear navigation.
Administrator / Faculty	Manage question banks, create/configure/schedule exams, manage relevant user status, and view reports.	Higher-privilege, trained user.

Role clarification: For access-control purposes, the current baseline uses two roles: **Student** and **Administrator**. Faculty users are treated as administrative users rather than as a third RBAC role. This preserves the Student-vs-Administrator access model in the original D2 while retaining D1's Administrator/Faculty terminology.

3.4 Operating Environment

Layer	Environment / technology
Client	Modern web browser; responsive web interface.
Frontend	React.js
Backend	Node.js + Express.js
Database	MongoDB
Authentication	JWT
Programming language	JavaScript
Development / version control	Git + GitHub
Deployment / containerization	Docker; CI/CD using Jenkins.

3.5 Design and Implementation Constraints

- The proposed implementation is web-based and uses the MERN stack specified in Deliverable 1.
- Objective questions are the supported automatically evaluated question type.
- The system does not include AI/webcam proctoring, live surveillance, subjective-answer evaluation, ERP integration or native mobile applications.
- Authentication is JWT-based and authorization is role-based.
- Testing and project evidence are expected to use the tools identified in the project plan.

3.6 Assumptions and Dependencies

- Users have access to a supported web browser and a functioning network connection.
- The MongoDB database service is available to the application.
- Students and administrators use valid registered credentials.
- Exam schedules, durations and question-selection criteria are configured by authorized administrative users.
- CI/CD, testing and static-analysis tools are available during the corresponding project activities.

4. External Interface Requirements

4.1 User Interfaces

Interface	Required logical behavior
Authentication	Registration/login forms; validation feedback; successful authentication establishes the user session.
Student Dashboard	Shows available exams, relevant status and navigation to exam/result/performance functions.
Exam Interface	Displays current question, remaining time, answer state, navigation controls and marked-for-review state.
Result / History	Shows completed attempt results, scores and performance information belonging to the authenticated student.
Admin Dashboard	Provides navigation to question bank, exam configuration/scheduling and reports.
Question Bank UI	Supports creation, editing, categorization and disabling of multiple-choice questions.
Exam Management UI	Supports duration, availability window and question-selection criteria.
Reports UI	Presents exam-wise and student-wise performance statistics to authorized administrators.

4.2 Software Interfaces

Interface	Input / output	Purpose
React.js ↔ REST API	JSON requests/responses	Client-server communication for authentication, question, exam, attempt and result operations.
Express.js ↔ MongoDB	Application data / database records	Persistence of users, questions, exams, attempts and results.
JWT ↔ secured API	Token in authenticated request	Authentication and role-based authorization.
Testing tools ↔ application	HTTP requests / test assertions	Functional, integration, security and API validation.

4.3 Communication Interfaces

- Client-server communication shall use HTTP/HTTPS with REST-style API endpoints.
- JSON shall be used for application-level request and response payloads.
- Authenticated requests shall carry a valid JWT according to the security requirements.
- HTTPS shall be used for production deployment to protect credentials and application data in transit.
- Auto-save and retry behavior shall be implemented through application/API communication so that a recoverable saved state is available after a temporary failure.

5. Analysis Models

5.1 Use Case Model

The use-case model covers the principal actors and interactions defined in the original D2 use-case list. The diagram below is the consolidated analysis model; detailed use-case records are provided in Section 6 and Appendix C.

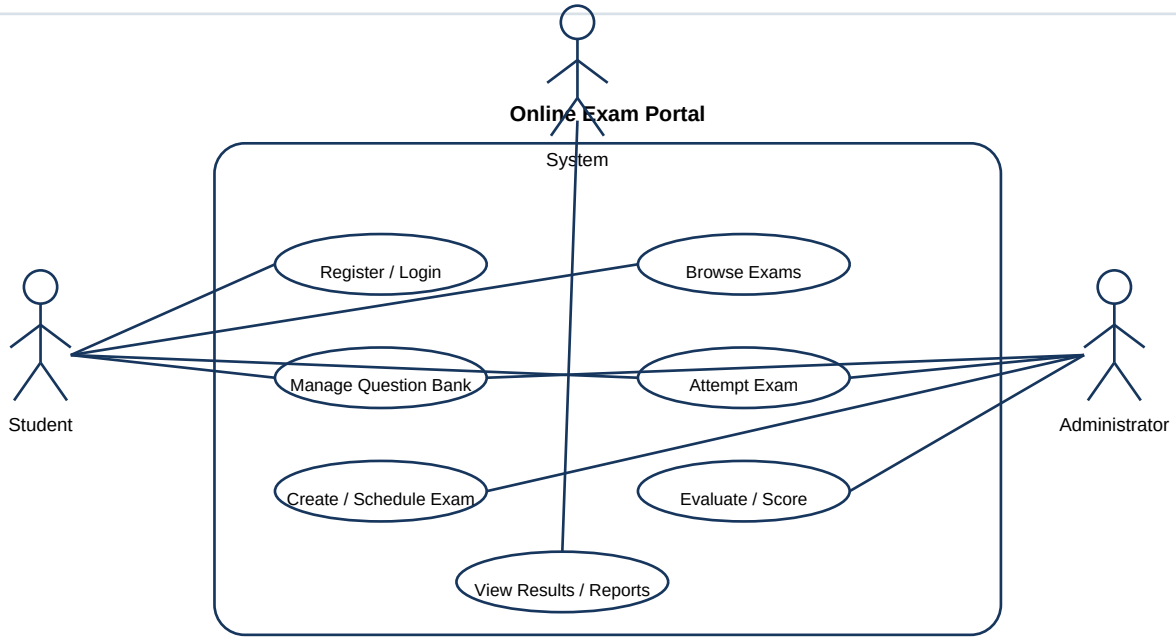

Figure 1. Consolidated use-case model for the Online Exam Portal.

5.2 Entity / Domain Model

The domain model below represents the core data objects needed to support the stated requirements. It is intentionally limited to entities directly required by the project scope.

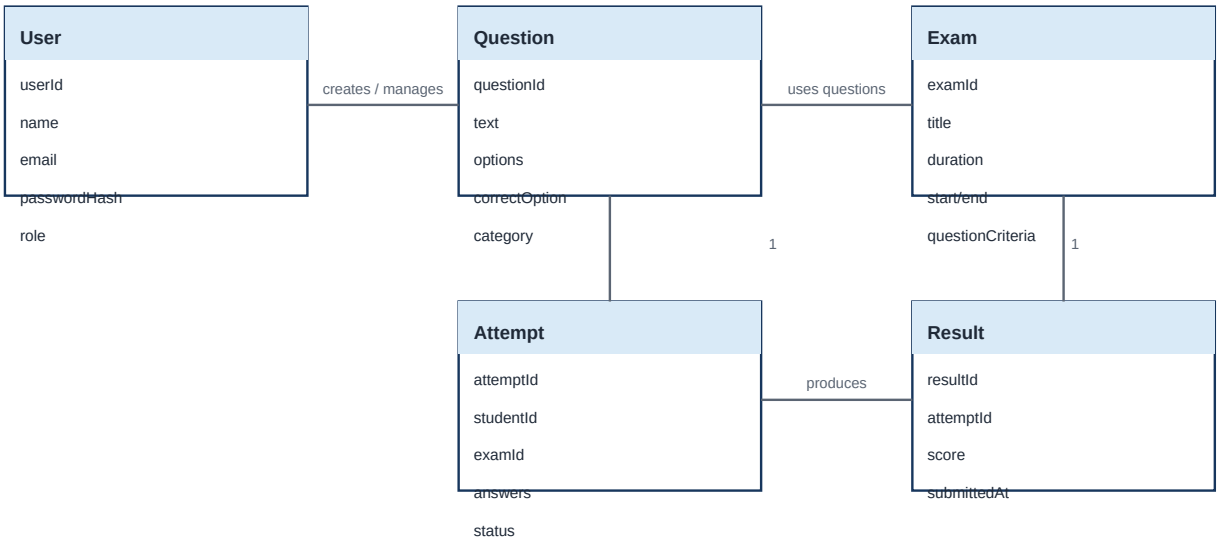

Figure 2. Core domain entities and relationships.

6. System Features and Functional Requirements

Each feature below includes its purpose, priority, stimulus/response behavior and the functional requirements that implement it. The requirements retain the original D2 FR identifiers so downstream traceability remains stable.

6.1 Authentication and Role-Based Access

Description: Provides registration/login and separates student and administrative privileges.
Priority: High

Stimulus / Response Sequences

Stimulus	System response
Valid registration/login	System validates credentials, creates/authenticates the user and establishes the authenticated session.
Duplicate/invalid credentials	System rejects the operation and reports the validation/authentication failure.
Unauthorized role access	System denies the request and does not expose privileged functionality.

Functional Requirements

ID	Priority	Requirement	Verification
FR-001	High	The system shall allow a user to register and login using a unique email address.	Functional test
FR-002	High	The system shall implement role-based access control (Student vs. Administrator).	Authorization tests

6.2 Question Bank Management

Description: Allows authorized administrators to maintain a reusable categorized bank of multiple-choice questions.
Priority: High

Stimulus / Response Sequences

Stimulus	System response
Administrator submits a valid question	System validates required fields and stores the question in the selected category.
Administrator edits/disables a question	System updates the question state while preserving consistency with existing exam/attempt rules.
Unauthorized student request	System rejects the request.

Functional Requirements

ID	Priority	Requirement	Verification
FR-003	High	The system shall allow administrators to create, edit, disable, and categorize multiple-choice questions.	Functional test

6.3 Exam Creation and Scheduling

Description: Allows administrators to configure examinations using duration, valid time window and question-selection criteria.
Priority: High

Stimulus / Response Sequences

Stimulus	System response
Administrator submits valid exam configuration	System validates the configuration and makes the exam available according to its schedule.
Student requests an exam outside the valid window	System does not permit the attempt to start.
Attempt already active	System prevents a second concurrent active attempt for the same student and exam.

Functional Requirements

ID	Priority	Requirement	Verification
FR-004	High	The system shall allow administrators to create an exam specifying duration, valid time window, and question selection criteria.	Functional test
FR-005	High	The system shall randomly select and shuffle questions from the designated question bank for each student attempt.	Unit + System test
FR-006	High	The system shall display available exams to the authenticated student based on the current valid time window.	Functional test

6.4 Student Exam Attempt

Description: Provides the timed examination workflow, navigation, marked-for-review state and auto-save.
Priority: High

Stimulus / Response Sequences

Stimulus	System response
Student starts an available exam	System creates an active attempt and displays the selected/shuffled questions with remaining time.
Student answers/navigates	System records the selected answer and permits movement between questions and marking for review.
Network/browser failure	Previously auto-saved state remains recoverable according to the reliability requirement.
Time expires	System automatically submits the active attempt.

Functional Requirements

ID	Priority	Requirement	Verification
FR-007	High	The system shall provide a timed interface for exam attempts, showing time remaining.	System test
FR-008	Medium	The system shall allow students to navigate between questions, and mark questions for review.	Usability test
FR-009	High	The system shall automatically save the student's selected answers during the exam.	Integration test
FR-010	High	The system shall automatically submit the exam when the allocated time limit expires.	Integration test
FR-011	High	The system shall prevent multiple concurrent active attempts for the same exam by the same student.	Negative integration test

6.5 Evaluation and Result Generation

Description: Evaluates objective questions after submission and stores the completed result.
Priority: High

Stimulus / Response Sequences

Stimulus	System response
Attempt submitted	System evaluates objective responses, calculates the score and stores the final result record.
Unattempted question	System assigns 0 marks; configurable negative marking applies according to the exam rule.
Submitted attempt	System does not permit further modification/resumption.

Functional Requirements

ID	Priority	Requirement	Verification
FR-012	High	The system shall automatically evaluate objective questions and calculate the total score upon exam submission.	Unit + System test
FR-013	High	The system shall generate and store a final result record for each completed attempt.	Integration test

6.6 Student Performance and Administrator Reporting

Description: Provides students with their own performance information and administrators with operational/exam analytics.
Priority: Medium

Stimulus / Response Sequences

Stimulus	System response
Student requests history	System displays only the authenticated student's completed attempt history, scores and performance information.
Administrator requests report	System displays authorized exam-wise and student-wise performance statistics.

Functional Requirements

ID	Priority	Requirement	Verification
FR-014	Medium	The system shall allow students to view their attempt history, scores, and performance dashboard.	Functional test
FR-015	Medium	The system shall provide administrators with reports detailing exam-wise statistics and student-wise performance.	Functional test

7. Nonfunctional, Security and Business Requirements

7.1 Performance Requirements

ID	Category	Measurable requirement	Verification
NFR-001	Performance	Exam-attempt API endpoints shall respond within 500 ms for 95% of requests under the defined test load.	Performance test
NFR-002	Availability	The application shall target 99.9% availability during active exam windows when deployed in the intended test/deployment environment.	Operational review
NFR-003	Usability	A student shall be able to start an available exam in no more than 3 clicks from the dashboard.	Usability walkthrough
NFR-004	Reliability	In case of browser crash or network failure, the auto-saved state shall be restorable within 1 minute of reconnection.	System test
NFR-005	Maintainability	Backend business logic for evaluation shall be isolated and covered by at least 80% automated unit tests.	Code review
NFR-006	Testability	The random question selector and evaluation engine shall be testable independently of the web framework.	Unit / integration test

7.2 Safety Requirements

No physical safety function is identified for this software-only examination portal. The relevant safeguard requirements are therefore expressed through reliability, data integrity, access control and prevention of invalid examination state transitions. The system shall prevent actions that could corrupt a submitted attempt or expose restricted examination content.

7.3 Security Requirements

ID	Security requirement	Validation approach
SEC-001	Passwords shall be securely hashed using bcrypt before storage.	Code / configuration review
SEC-002	Role-based APIs shall strictly reject requests from users without appropriate privileges.	Authorization tests
SEC-003	All user-supplied inputs shall be validated and sanitized to prevent NoSQL injection and XSS.	Negative / security tests
SEC-004	JWT tokens shall have a reasonable expiration time and be validated on every secured request.	Security test
SEC-005	Exam questions shall not be exposed in API responses to students prior to the start of their active attempt.	Design review
SEC-006	The system shall log failed authentication attempts and unauthorized API access attempts.	System test

7.4 Business / Domain Rules

- A student cannot start an exam outside its configured valid time window.
- An exam attempt cannot be modified or resumed once it has been submitted, either manually or automatically.
- An exam configuration cannot be modified if any student has already started an attempt for that exam.
- Only authorized administrative users can create or modify questions and exams.
- The evaluation engine assigns 0 marks for unattempted questions; negative marking is configurable per exam.

7.5 Software Quality Attributes

Attribute	Requirement / interpretation	Evidence
Correctness	Evaluation and randomization logic shall be independently testable and validated.	Unit + integration tests
Reliability	Auto-saved state shall be recoverable after browser/network failure within the specified target.	System/failure tests
Maintainability	Evaluation business logic shall be isolated and maintainable; target unit-test coverage is at least 80%.	Code review + coverage
Testability	Core selection/evaluation components shall not depend directly on the web framework.	Unit/integration tests
Usability	Student exam start shall require no more than 3 clicks from the dashboard.	Task-based walkthrough
Security	Authentication, authorization, input validation and protected question access shall be enforced.	Security tests

8. Other Requirements

8.1 Database Requirements

- MongoDB shall persist user, question, exam, attempt and result data required by the system.
- Stored data shall preserve the relationships necessary to reconstruct completed attempts and results.
- Unique email addresses shall be enforced for registered users.
- Question, exam and attempt records shall support the business rules defined in Section 7.

8.2 Data Retention and Integrity

- Completed attempt and result records shall be retained for the project-defined retention period.
- Submitted attempts shall be immutable through normal application operations.
- Auto-saved responses shall be associated with the corresponding active attempt.

8.3 Deployment Requirements

- The application shall be containerizable using Docker.
- CI/CD shall be demonstrated using Jenkins as specified in Deliverable 1.
- Git/GitHub shall be used for version control and project history.
- The deployed environment shall provide access to the backend and MongoDB services required by the application.

8.4 Localization

The initial implementation targets English-language user interfaces. No multilingual requirement is included in the current project scope.

8.5 Reuse and Extensibility

The categorized question-bank and modular evaluation design should allow additional objective-question categories or scoring configurations to be introduced later without changing the core authentication and attempt workflow. Such extensions are outside the current baseline unless explicitly added in a future revision.

9. Validation Specification – Initial Planning

Validation will verify that the implemented system satisfies the functional, nonfunctional, security and reliability requirements. The following plan preserves the validation categories from the original D2 and expands them into concrete evidence targets.

Requirement category	Planned validation	Example / evidence
Functional	Black-box functional tests	Valid and invalid registration, question creation, exam configuration, exam attempt, submission and result workflows.
Security	Authentication, authorization and negative security testing	Student attempts an administrator endpoint; invalid/expired JWT; malicious input; restricted question access.
Performance	Load and response-time testing	Concurrent exam-attempt requests; verify NFR-001 target.
Usability	Task-based walkthrough	New student logs in and starts an available exam within the stated click target.
Reliability	Failure/retry scenarios	Network timeout, browser failure and auto-save recovery.
Unit	Automated unit tests and coverage	Score calculation and random question-selection logic; verify NFR-005 target.
Integration	API/database/workflow tests	Attempt creation, auto-save, auto-submit, evaluation and result persistence.
Static analysis	Automated static analysis	SonarQube or similar report for maintainability/security issues.
CI/CD	Pipeline validation	Jenkins build/test pipeline evidence.

9.1 Acceptance-oriented Test Targets

ID	Acceptance target
FR-001 / FR-002	Valid authentication succeeds; duplicate/invalid credentials fail; unauthorized role access is rejected.
FR-003 / FR-004	Administrator can maintain questions and configure an exam with valid scheduling/selection criteria.
FR-005	Independent attempts demonstrate question selection/shuffling behavior according to the configured question bank.
FR-006–FR-011	Only eligible students can start; timer, navigation, marked-for-review, auto-save, auto-submit and concurrent-attempt prevention behave as specified.
FR-012 / FR-013	Submitted objective attempts produce the expected score and a stored final result.
FR-014 / FR-015	Students see only their own performance information; administrators can access the defined reports.
NFR-001–NFR-006	Performance, availability target, usability, recovery, coverage and testability targets are evaluated with the evidence types specified above.
SEC-001–SEC-006	Security controls are verified through code/configuration review, authorization tests, negative tests and system/design review.

10. Project Plan

The work-package plan below is aligned with the six-sprint plan in Deliverable 1. Work packages are treated as implementation streams, while the sprint mapping shows when each stream is expected to contribute.

WP	Tasks	Owner	Effort (person-days)	Output	Sprint alignment
WP1	Requirements + use cases	All	4	SRS / requirements baseline	Sprint 1
WP2	Architecture + UML	Samarth / Saikavin	4	Architecture / design document	Sprint 1–2
WP3	Authentication + security	Saikavin	6	Working authentication/security features	Sprint 1 + Sprint 5
WP4	Question bank	Samarth	6	Working question-bank feature	Sprint 2
WP5	Exam engine	Samrudh	7	Working exam/attempt feature	Sprint 2–3
WP6	Student dashboard + performance	Satyam	7	Working dashboard/performance feature	Sprint 5
WP7	Testing + CI/CD + security validation	All	6	Test/validation evidence	Sprint 5–6
WP8	Final validation + report	All	4	Final report and release evidence	Sprint 6

10.1 Sprint-Level Plan

Sprint	Planned output	Acceptance evidence
Sprint 1	Project setup, requirements baseline, JWT authentication (register/login).	User stories, Git repository and working login/register.
Sprint 2	Question-bank CRUD and administrator exam creation.	Demonstration and unit tests.
Sprint 3	Student exam-attempt engine: randomization, navigation and auto-save.	Integration tests and pull requests.
Sprint 4	Evaluation/scoring engine and result generation.	System demonstration and test evidence.
Sprint 5	Student performance dashboard, administrator analytics and security hardening.	Validation and coverage report.
Sprint 6	System validation, CI/CD and final release.	Final report, deployment and demonstration.

10.2 Product Ownership

Member	Owned functional feature	Secondary / qualitative responsibility
Samarth Kotian	Question Bank & Exam Evaluation — question creation/editing, categorization, randomization, objective evaluation, scoring and result generation.	Database architecture and data consistency.
Saikavin S	Authentication & Administration — registration/login, JWT authentication, RBAC, administrator functions and user/system management.	Security validation and access-control testing.
Samrudh C Patil	Exam Management & Attempt Processing — exam creation/configuration, scheduling, attempt lifecycle, auto-save and automatic submission.	REST API reliability, validation and error handling.
Satyam Goyal	Student Examination & Performance — examination interface, question navigation, result viewing, history and performance dashboard.	Usability and responsive student experience.

11. Tool Stack and Evidence

Activity	Tool / technology	Evidence to submit
Frontend	React.js	Source code, UI demonstration
Backend	Node.js + Express.js	Source code, API demonstration
Database	MongoDB	Schema/data evidence and integration tests
Authentication	JWT	Authentication/authorization test evidence
Programming language	JavaScript	Repository/code
Version control	Git + GitHub	Repository URL and commit history
Agile planning	Jira or GitHub Projects	Backlog, user stories, sprint board
CI/CD	Jenkins	Pipeline screenshot/log
Containerization	Docker	Dockerfile and build/run evidence
Testing	Jest / Testing Library or appropriate JavaScript testing tools	Test results and coverage
Static analysis	SonarQube or similar	Analysis report
API testing	Postman	Collection and results
Design	Draw.io / StarUML	UML diagrams

Appendix A — Glossary

Term	Definition
Administrator	Administrative user with privileged access to question-bank, exam-management and reporting functions. Faculty users are treated under this administrative role in the current RBAC baseline.
Attempt	A student's active or completed participation in a specific examination.
Auto-save	Automatic persistence of selected answers during an active exam attempt.
Auto-submit	Automatic submission of an active exam when the configured time limit expires.
Exam	A configured and scheduled collection of questions that a student may attempt during its valid time window.
JWT	JSON Web Token used by the proposed authentication mechanism for authenticated requests.
Question Bank	Reusable, categorized collection of multiple-choice questions maintained by authorized administrators.
RBAC	Role-Based Access Control; permissions are determined by the user's role.
Result	The stored outcome of a completed exam attempt, including the calculated score.
Marked for Review	A student interface state used to flag a question for revisiting before submission.
Availability Window	Configured period during which an exam may be started by eligible students.

Appendix B — Field Layouts and Report Requirements

The following field layouts define the minimum data objects implied by the requirements. Exact MongoDB schema syntax may be refined during design without changing the externally visible requirements.

B.1 User

Field	Type	Required	Description
userId	ObjectId/String	Yes	Unique user identifier.
name	String	Yes	User display/name value.
email	String	Yes	Unique login identifier.
passwordHash	String	Yes	Securely hashed password; plaintext password shall not be stored.
role	Enum	Yes	Student or Administrator.
status	Enum	Yes	Account state used for administrative user management.

B.2 Question

Field	Type	Required	Description
questionId	ObjectId/String	Yes	Unique question identifier.
text	String	Yes	Question statement.
options	Array	Yes	Multiple-choice options.
correctOption	String/Index	Yes	Correct option used by evaluation.
category	String	Yes	Question-bank category.
status	Enum	Yes	Active/disabled state.

B.3 Exam

Field	Type	Required	Description
examId	ObjectId/String	Yes	Unique exam identifier.
title	String	Yes	Exam title.
duration	Number	Yes	Configured duration.
startTime	DateTime	Yes	Start of valid availability window.
endTime	DateTime	Yes	End of valid availability window.
questionCriteria	Object/Array	Yes	Rules for selecting questions from the bank.
negativeMarking	Number/Boolean	No	Configurable negative-marking rule.

B.4 Attempt

Field	Type	Required	Description
attemptId	ObjectId/String	Yes	Unique attempt identifier.
studentId	Reference	Yes	Owning student.
examId	Reference	Yes	Exam being attempted.
questionOrder	Array	Yes	Selected/shuffled question order.
answers	Array/Object	Yes	Saved student responses.
status	Enum	Yes	Active, submitted or equivalent lifecycle state.
startedAt	DateTime	Yes	Attempt start time.
submittedAt	DateTime	No	Submission time when completed.

B.5 Result

Field	Type	Required	Description
resultId	ObjectId/String	Yes	Unique result identifier.
attemptId	Reference	Yes	Associated completed attempt.
score	Number	Yes	Calculated score.
submittedAt	DateTime	Yes	Completion/submission timestamp.

B.6 Report Requirements

Report	Required information	Access
Student Performance	Own attempt history, scores and performance information.	Authenticated student; own records only.
Exam-wise Statistics	Performance statistics associated with an examination.	Administrator.
Student-wise Performance	Performance information organized by student.	Administrator.

Appendix C — Requirement Traceability Matrix

This matrix establishes the baseline relationship from requirements to analysis/architecture modules, design components and planned tests. Code identifiers are intentionally expressed as implementation placeholders until the repository structure is finalized.

Requirement	Analysis / architecture	Design / module	Planned test / evidence
FR-001	Authentication	Auth controller/service	TC-AUTH-01 to TC-AUTH-03
FR-002	RBAC	Authorization middleware	TC-SEC-01 to TC-SEC-03
FR-003	Question Bank	Question controller/service	TC-QB-01 to TC-QB-05
FR-004	Exam Management	Exam controller/service	TC-EXAM-01 to TC-EXAM-04
FR-005	Exam Engine	Random selection/shuffle service	TC-EXAM-05; unit + system evidence
FR-006	Exam Discovery	Student dashboard/API	TC-EXAM-06
FR-007	Attempt Engine	Timed attempt UI/service	TC-ATT-01
FR-008	Attempt Engine	Navigation/review state	TC-ATT-02
FR-009	Attempt Engine	Auto-save service/API	TC-ATT-03; recovery evidence
FR-010	Attempt Engine	Auto-submit service	TC-ATT-04
FR-011	Attempt Engine	Attempt lifecycle constraint	TC-ATT-05
FR-012	Evaluation	Evaluation service	TC-EVAL-01 to TC-EVAL-04
FR-013	Results	Result persistence/service	TC-RES-01
FR-014	Student Performance	History/dashboard module	TC-RES-02
FR-015	Reporting	Admin reporting module	TC-REP-01 to TC-REP-03
NFR-001	API / deployment architecture	API endpoints	PT-01 performance test
NFR-002	Deployment architecture	Runtime/availability setup	AV-01 operational review
NFR-003	UI architecture	Student dashboard/exam flow	UT-01 usability walkthrough
NFR-004	Attempt/recovery architecture	Auto-save/retry mechanism	REL-01 failure/retry test
NFR-005	Application architecture	Evaluation service	COV-01 coverage report
NFR-006	Application architecture	Randomizer/evaluation modules	TEST-01 independent unit/integration tests
SEC-001	Authentication architecture	Password hashing	SEC-01 code/config review
SEC-002	Security architecture	RBAC middleware	SEC-02 authorization tests
SEC-003	API/security layer	Validation/sanitization	SEC-03 negative tests
SEC-004	Authentication architecture	JWT middleware	SEC-04 token validation tests
SEC-005	Exam/attempt architecture	Protected question API	SEC-05 design/API review
SEC-006	Security/audit architecture	Authentication/access logging	SEC-06 system test

Appendix C.1 D1 Objective-to-D2 Traceability

Deliverable 1 objective	Deliverable 2 coverage
Secure role-based access using JWT	FR-001–FR-002; SEC-001–SEC-006
Reusable categorized question bank	FR-003; Question Bank feature
Create/configure/schedule exams	FR-004; Exam Management feature
Randomize question order/selection	FR-005; randomization test plan
Timed exam with navigation	FR-006–FR-011
Automatic objective evaluation/results	FR-012–FR-013
Student dashboard + administrator analytics	FR-014–FR-015
Agile development on MERN stack	Sections 3, 10 and 11

Appendix D — Revision History and Change Summary

Version	Date	Change
1.0	2026-09-07	Completed Deliverable 2 SRS structure; added overall description, external interfaces, analysis models, system features, other requirements, glossary, field layouts and traceability; aligned project plan with D1 sprints; clarified Student/Administrator RBAC and D1 user-management wording; retained original FR/NFR/SEC identifiers.

Final consistency check

- Project title, group, project ID, team members and MERN technology stack match Deliverable 1.
- D1 target users and objectives are represented in the D2 requirements.
- All ten D1 functional features are covered by the D2 functional requirements.
- D1 out-of-scope items remain out of scope.
- Original D2 FR/NFR/SEC identifiers are preserved for continuity.
- D2 now contains the required SRS sections and appendices identified in the supplied course guidelines.